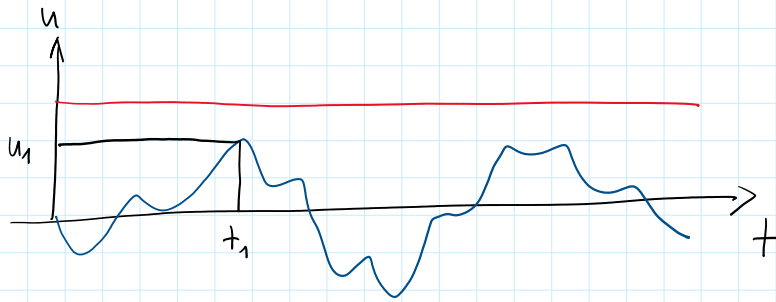
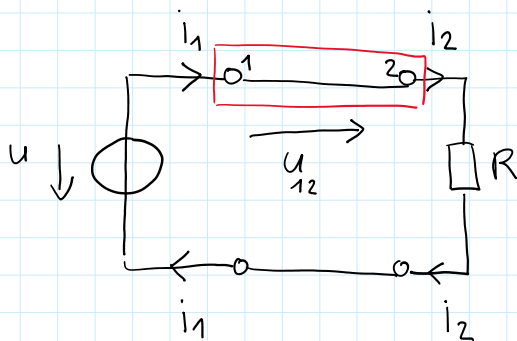


68 - 86 - 17 - 50



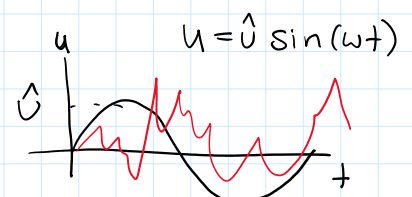
NOTATION

 U, I, P Gleichgrößen DC (direct current) $u, i, p, P(t)$ Wechselgrößen AC (alternating current)DC $u_{12} = \text{const.}$ $i_1 = i_2$ $\rightarrow$ ZP-Beding. ✓

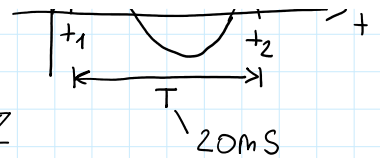
AC

 $v < \text{Lichtgeschwindigkeit } c$ $i_1 \neq i_2$ $\rightarrow$ ZP-Beding. ✗QUASISTATIONÄRE GRÖßEN $\lambda = \frac{c}{f} \gg \text{Geometrie} \rightarrow \text{ZP-Beding.} \checkmark$ 50 Hz $\rightarrow \lambda = 5.995,849 \text{ km}$ 2,4 GHz $\rightarrow \lambda = 12,491 \text{ cm}$ $c = 299.792.458 \text{ m/s}$ KLASSIFIZIERUNG FÜR GRÖßEN

- + analog - digital
- + deterministisch - stochastisch
- + ein- und mehrdimensional
- + konstant - quasistationär - instationär

PERIODISCH ZEITABHÄNGIGE GRÖßEN

PERIODISCH ZEITABHÄNGIGE GRÖßEN



$$x(t) = x(t + n \cdot T) \quad n = 0, 1, 2, \dots \quad n \in \mathbb{Z}$$

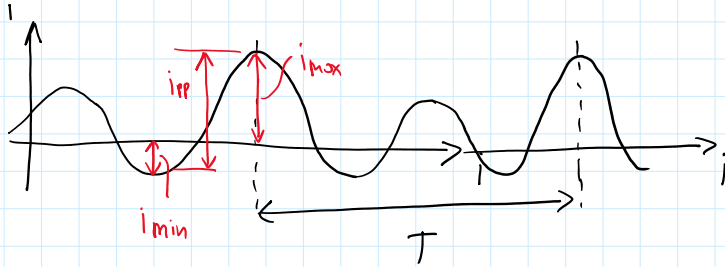
Frequenz f $[f] = \left(\frac{1}{s}\right) = \text{Hz}$

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

$$T = \frac{1}{f}$$

Kreisfrequenz ω $[\omega] = \frac{1}{s}$

$$\omega = 2\pi f \quad (\rightarrow \sin \text{ formige Fkt.})$$



$i_{\min}$ Minimumwert

$i_{\max}$ Maximumwert

i_{pp} Schwingbreite

$\uparrow$ Scheitelwert

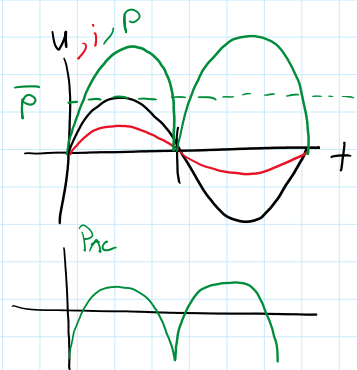
$\max(|i_{\max}|, |i_{\min}|)$

GLEICHWERT

$$\bar{u} = \frac{1}{T} \int_{t_0}^{t_0+T} u \, dt \quad (= \text{arithmetisches Mittelwert})$$

$$P = \frac{1}{T} \int p(t) \, dt \quad \text{Wirkleistung}$$

$$u = \bar{u} + u_{AC}$$



EFFEKTIVWERT RMS

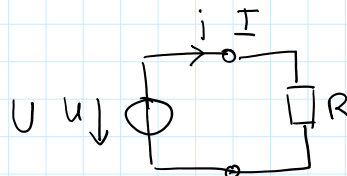
$$P(t) = R i^2$$

$$dW = P(t) dt = R i^2 dt$$

$$W_T = \int_0^T dW = R \int_0^T i^2 dt$$

$$R I^2 \cdot T = R \int_0^T i^2 dt$$

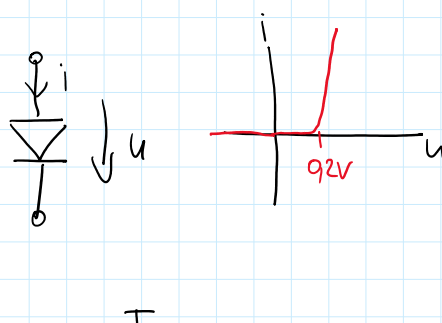
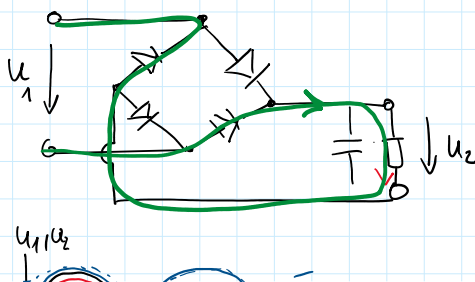
$$I = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

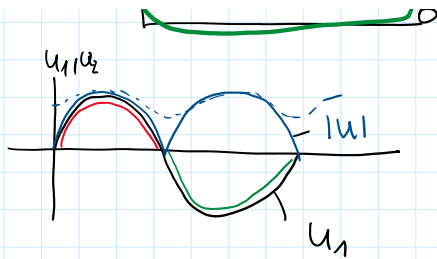


$$\underbrace{R I^2 \cdot T}_{DC}$$

$$U = \sqrt{\frac{1}{T} \int_0^T u^2 dt}$$

GLEICHRICHTWERT



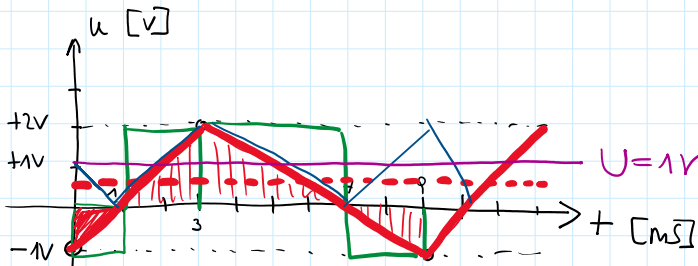


$$|\bar{u}| = \frac{1}{T} \int_0^T |u_1| \cdot dt$$

VERHÄLTNISSZAHLEN

Siehe Klaus

BSP E10



$$T = 9 \text{ ms} \rightarrow f = \frac{1}{T} = 111 \text{ Hz}$$

$$u_{\min} = -1 \text{ V} \quad u_{\max} = +2 \text{ V} \quad u_{pp} = 3 \text{ V}$$

$$\hat{u} = \max(|u_{\min}|, |u_{\max}|) = 2 \text{ V}$$

$$u_1 = k_1 t + d_1 \quad u_1(0) = -1 = k_1 \cdot 0 + d_1$$

$$u_1\left(\frac{3}{1000}\right) = 2 = k_1 \cdot \frac{3}{1000} + d_1$$

$$\boxed{u_1 = \frac{1000 \text{ V}}{\text{s}} t - 1 \text{ V}} \leftarrow d_1 = -1 \text{ V}, k_1 = 1000 \text{ V/s}$$

$$u_1 = 0 \rightarrow t_{10} = 1 \text{ ms}$$

$$u_2 = k_2 t + d_2$$

$$u_2\left(\frac{3}{1000}\right) = 2 \text{ V} = k_2 \cdot \frac{3}{1000} + d_2$$

$$u_2\left(\frac{9}{1000}\right) = -1 \text{ V} = k_2 \cdot \frac{9}{1000} + d_2$$

$$\boxed{u_2 = -500 \frac{\text{V}}{\text{s}} t + 3,5 \text{ V}}$$

$$d_2 = 3,5 \text{ V} \quad k_2 = -500 \text{ V/s}$$

$$u_2 = 0 \rightarrow t_{20} = 7 \text{ ms}$$

GLEICHWERT

$$\begin{aligned} \bar{u} &= \frac{1}{T} \int_0^T u \, dt = \frac{1000}{9} \text{ s} \left(-1 \text{ V} \cdot \frac{1}{1000} \text{ s} \cdot \frac{1}{2} + 2 \text{ V} \cdot \frac{2}{1000} \text{ s} \cdot \frac{1}{2} + 2 \text{ V} \cdot \frac{4}{1000} \text{ s} \cdot \frac{1}{2} - 1 \text{ V} \cdot \frac{2}{1000} \text{ s} \cdot \frac{1}{2} \right) = \\ &= \underline{\underline{\frac{1}{2} \text{ V}}} \end{aligned}$$

GLEICHRICHTWERT

$$|\bar{u}| = \frac{1}{T} \int_0^T |u| \, dt = \frac{1000}{9} \text{ s} (\dots) = \underline{\underline{\frac{5}{6} \text{ V}}}$$

EFFEKTIVWERT

$$u^2 = \frac{1}{T} \int_0^T u^2 \, dt = \frac{1}{9} \left(\int_0^{3 \text{ ms}} \dots + \int_{3 \text{ ms}}^{9 \text{ ms}} \dots \right) =$$

$$U^2 = \frac{1}{T} \int_0^T u^2 dt = \frac{1}{T} \left(\int_0^{3ms} u_1^2 dt + \int_{3ms}^{9ms} u_2^2 dt \right) =$$

$$u_1 = \frac{1000V}{s} t - 1V$$

$$u_1^2 = 1e6 t^2 - 2000t + 1$$

$$u_2 = -500 \frac{V}{s} t + 3,5V$$

$$u_2^2 = 250.000 t^2 - 3500t + \frac{49}{4}$$

$$\begin{aligned} \int_0^{3/1000} 1e6 t^2 - 2000t + 1 dt &= \left[1e6 \frac{t^3}{3} - 1000t^2 + t \right]_0^{3/1000} = \\ &= \frac{9}{1000} - \frac{9}{1000} + \frac{3}{1000} = \frac{3}{1000} \end{aligned}$$

$$\begin{aligned} \int_{3/1000}^{9/1000} 250000 t^2 - 3500t + \frac{49}{4} dt &= \left[\frac{250000}{3} t^3 - \frac{3500}{2} t^2 + \frac{49}{4} t \right]_{3/1000}^{9/1000} = \\ &= \frac{243}{4000} - \frac{567}{4000} + \frac{441}{4000} - \frac{9}{4000} + \frac{63}{4000} - \frac{147}{4000} = \frac{24}{4000} = \frac{6}{1000} \end{aligned}$$

$$U^2 = \frac{1000}{9} \left(\frac{3}{1000} + \frac{6}{1000} \right) = 1 V^2 \quad \underline{U = \sqrt{1V^2} = 1V}$$